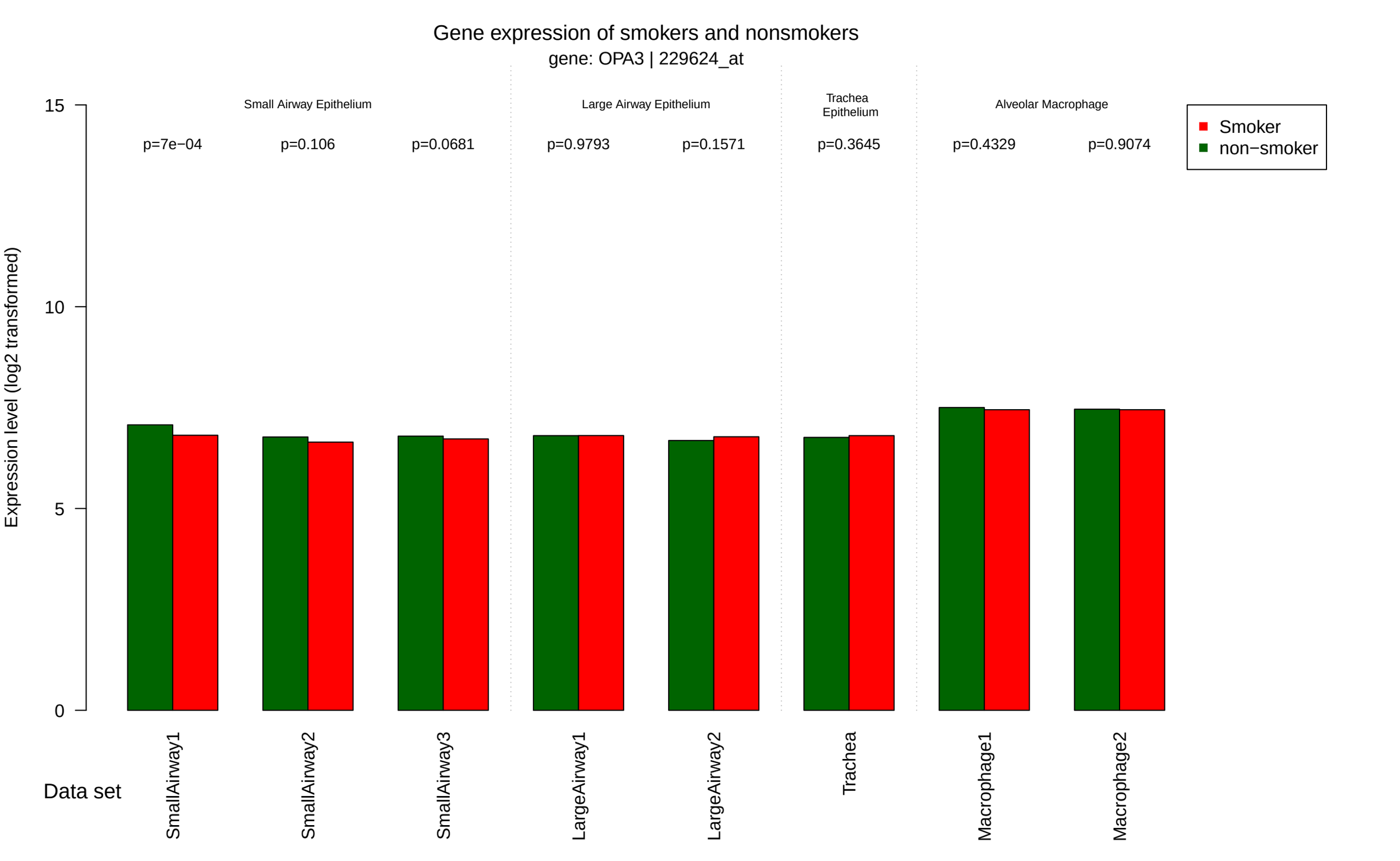
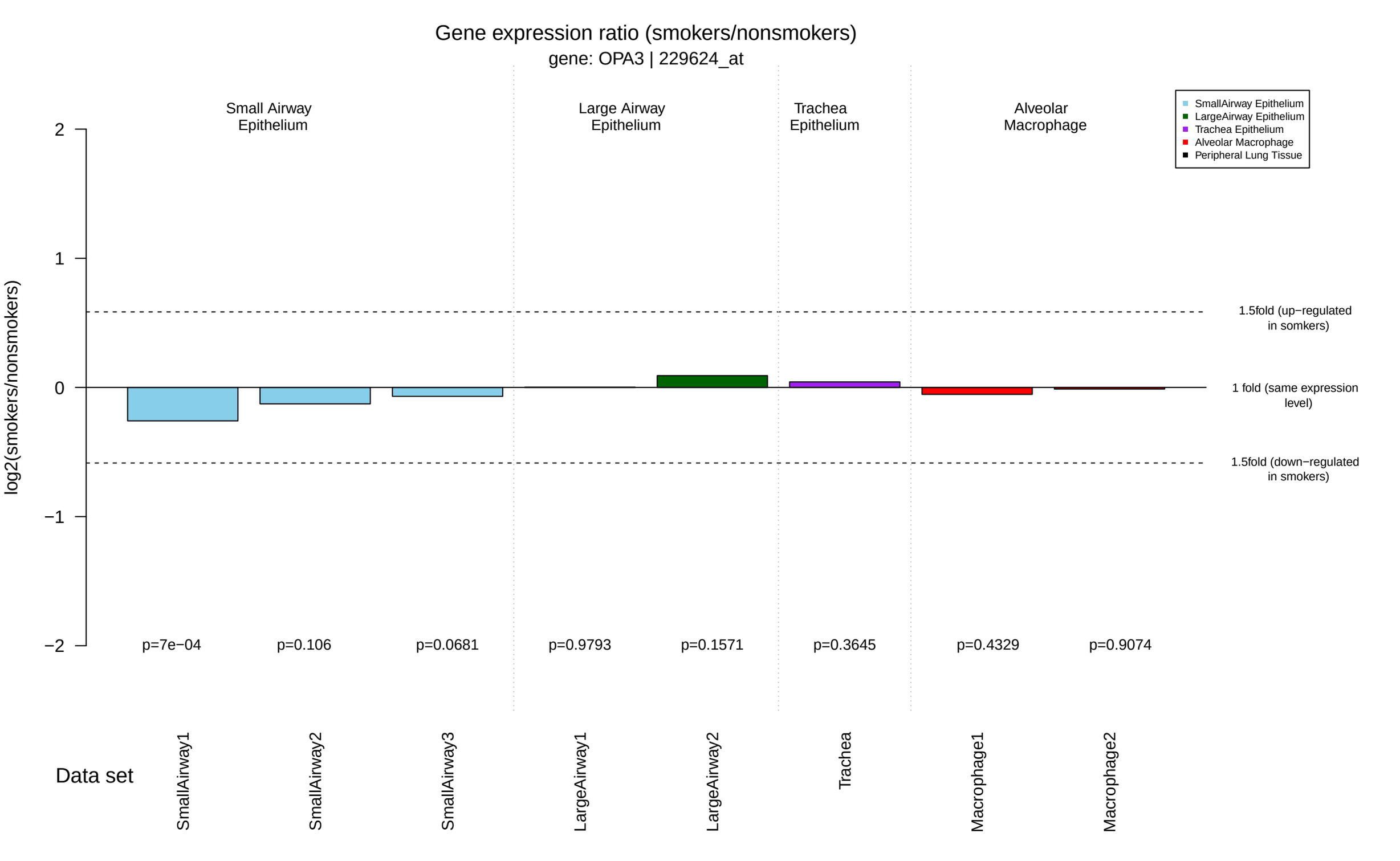
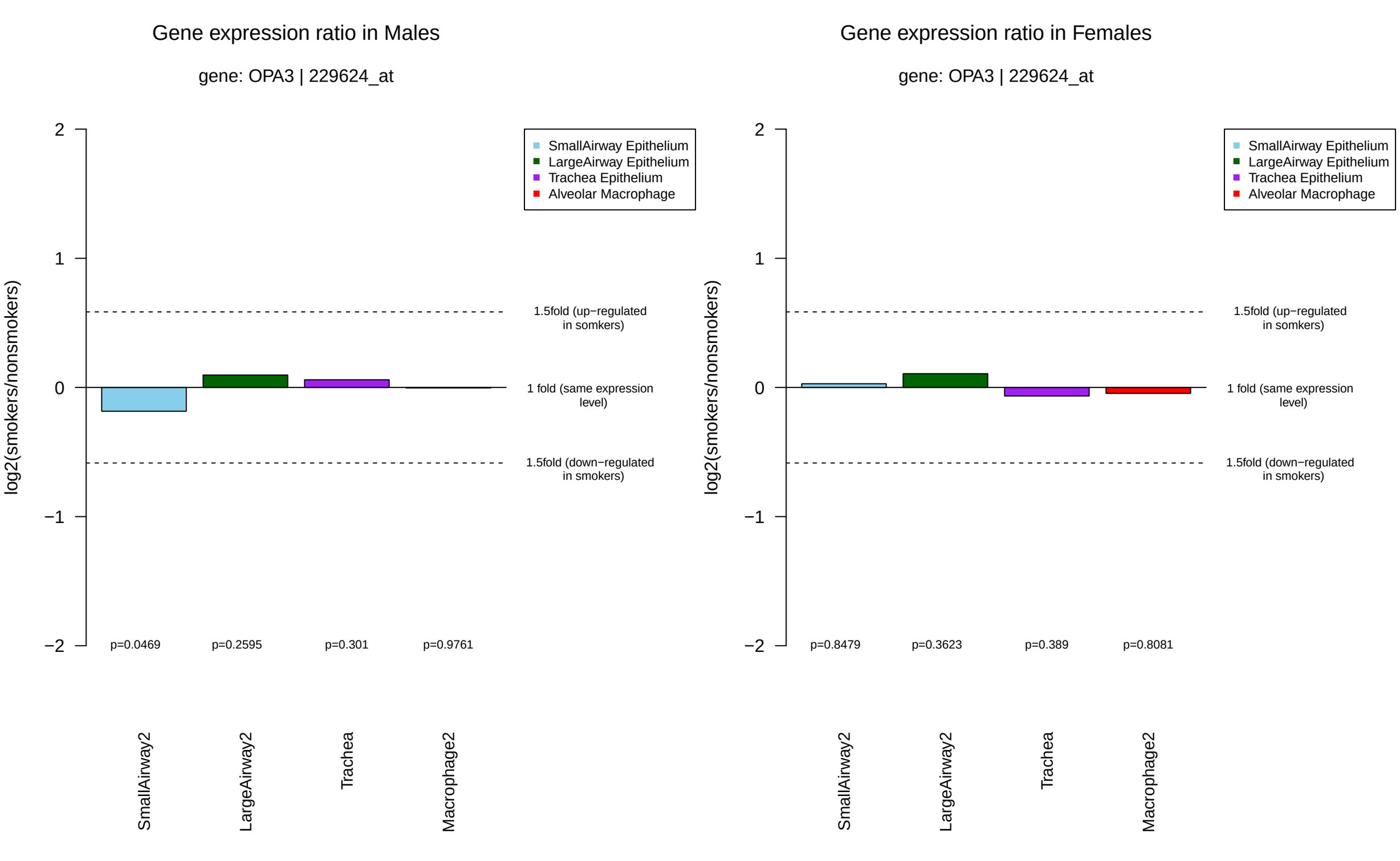
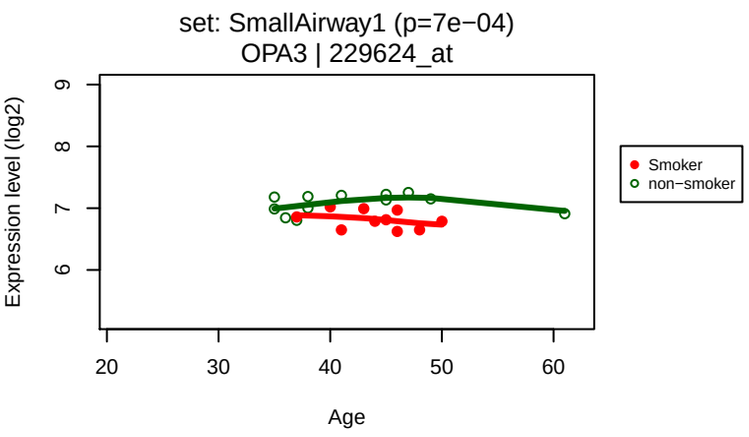
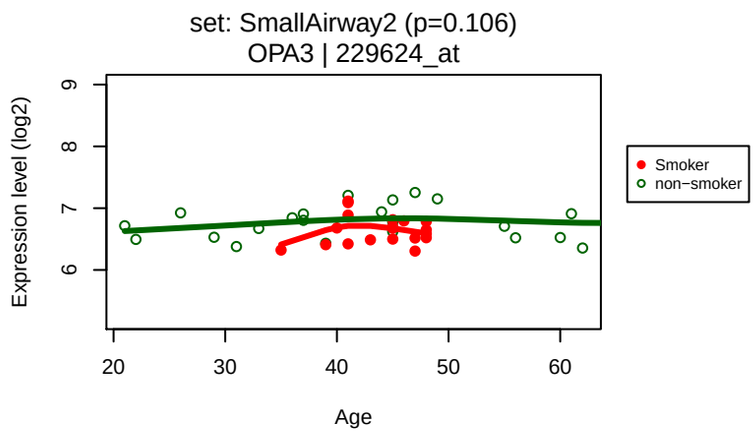
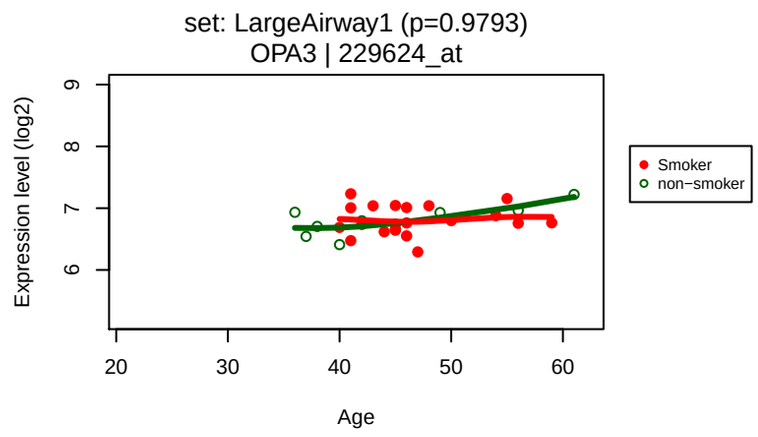

Gene expression of smokers and nonsmokers
gene: OPA3 | 229624_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=7e−04
p=0.106
p=0.0681
p=0.9793
p=0.1571
p=0.3645
p=0.4329
p=0.9074
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

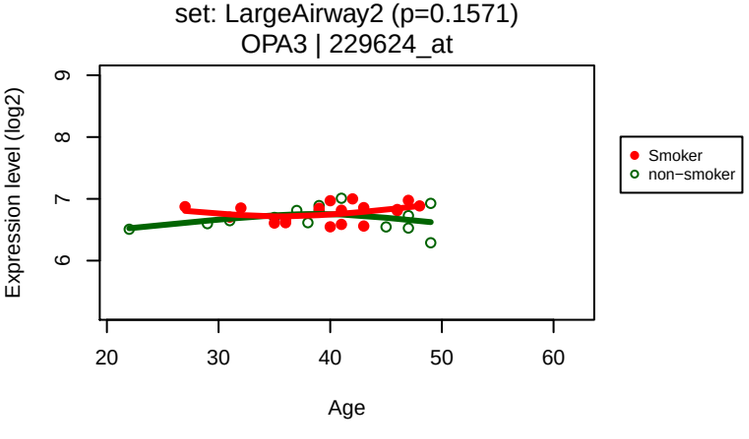

Gene expression ratio (smokers/nonsmokers)
gene: OPA3 | 229624_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=7e−04
p=0.106
p=0.0681
p=0.9793
p=0.1571
p=0.3645
p=0.4329
p=0.9074
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

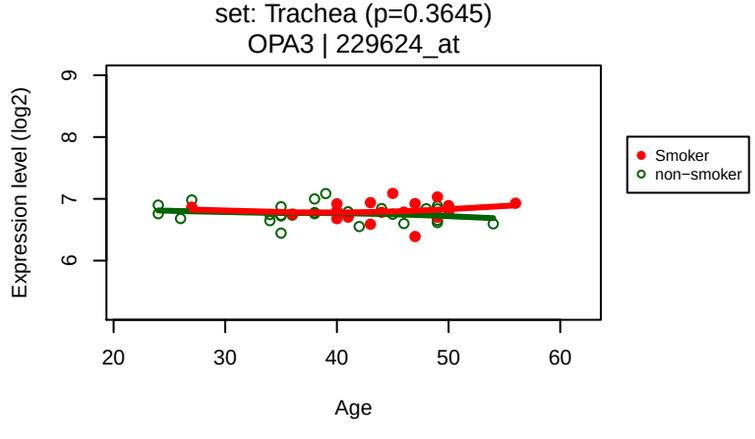

Gene expression ratio in Males
gene: OPA3 | 229624_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0469
p=0.2595
p=0.301
p=0.9761
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OPA3 | 229624_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8479
p=0.3623
p=0.389
p=0.8081
SmallAirway2
LargeAirway2
Trachea
Macrophage2

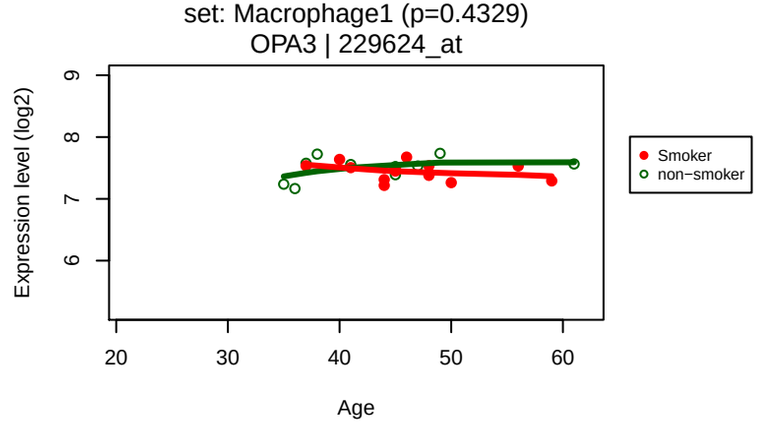

set: SmallAirway1 (p=7e−04)
OPA3 | 229624_at
Expression level (log2)
Age
Smoker
non−smoker

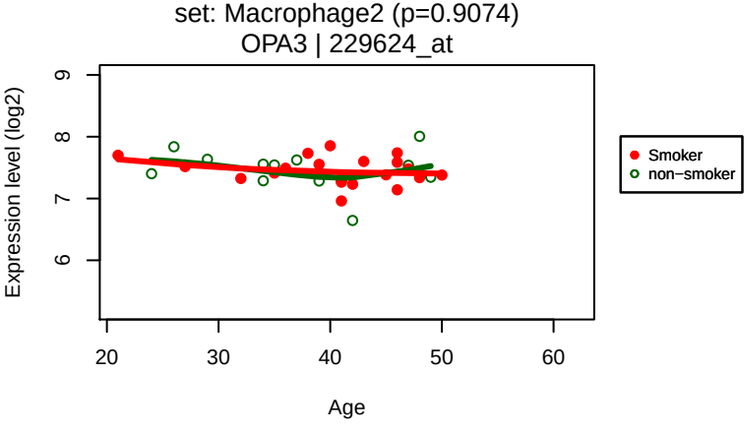

set: SmallAirway2 (p=0.106)
OPA3 | 229624_at
Expression level (log2)
Age
Smoker
non−smoker

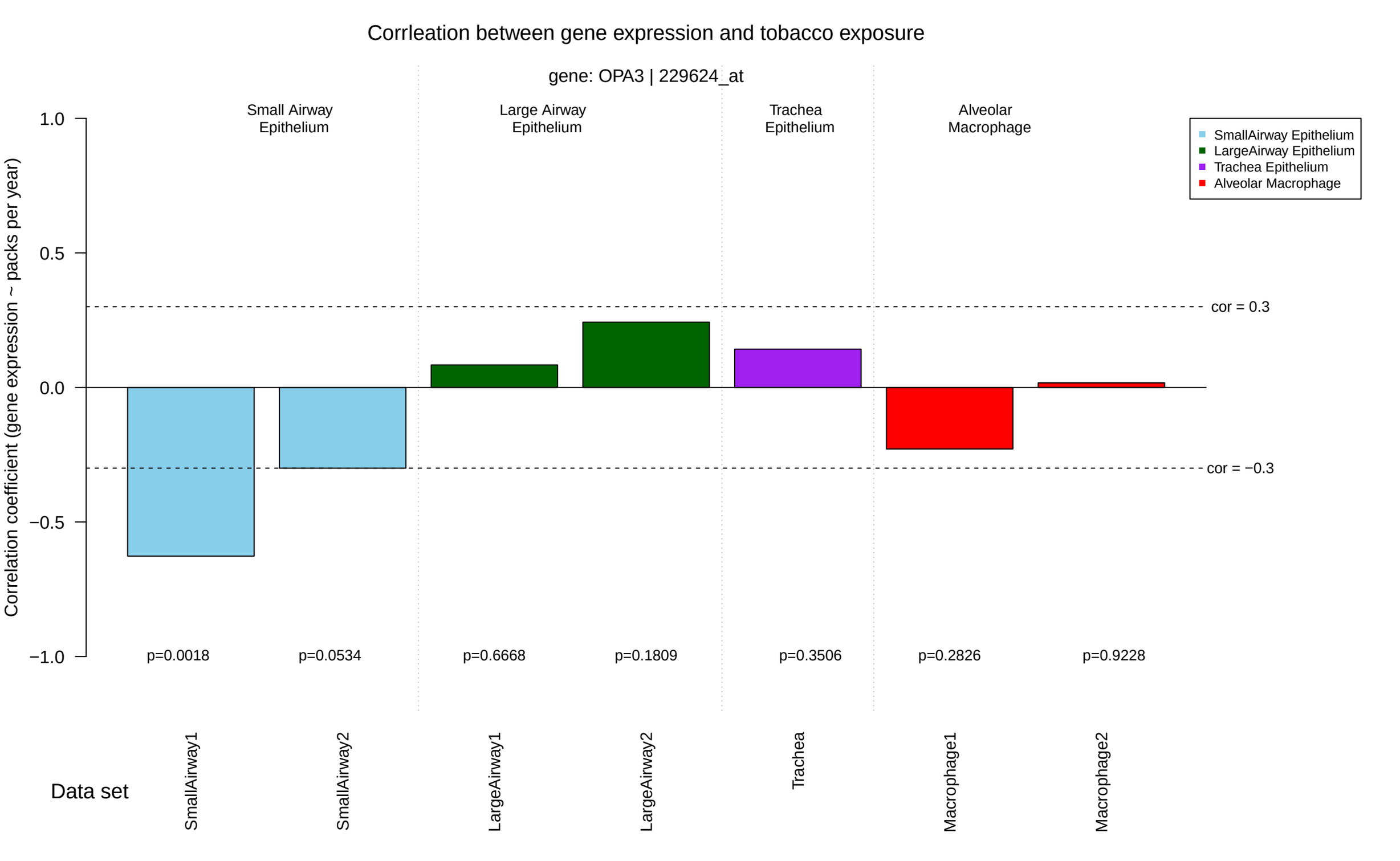

set: LargeAirway1 (p=0.9793)
OPA3 | 229624_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1571)
OPA3 | 229624_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3645)
OPA3 | 229624_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4329)
OPA3 | 229624_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9074)
OPA3 | 229624_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OPA3 | 229624_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0018
p=0.0534
p=0.6668
p=0.1809
p=0.3506
p=0.2826
p=0.9228
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2